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Project Baseball: Chicago Cubs vs. Baltimore Orioles

Statement of the problem

Our project is motivated by the idea of simulating a baseball series between two MLB teams using various modeling techniques. To do this, we aim to simulate a three-game series between the Chicago Cubs and the Baltimore Orioles through Markov Chain analysis, using only individual batting statistics like singles, doubles triples, HRs, strikeouts, walks, etc. for each team. Batting statistics were gathered through ESPN's 2025 regular season statistics through all 162 games in the season. The winner is determined by whichever team accumulates more wins by the end of the three games played.

Mathematical model

Here are the definitions of all our batting statistics used:

- Plate Appearances: Total # of times a batter comes to the plate.
- AB: Total # of times a player comes to the plate and faces a pitch. Excludes walks, hit by pitches, and errors.
- Walks: When the batter gets a free advance to first base due to a pitcher throwing four balls outside strike zone.
- Singles: A hit where the batter reaches first base.
- Doubles: A hit where the batter reaches second base.
- Triples: A hit where the batter reaches third base.
- Home Runs: A hit where the batter gets through all bases and scores.
- Hit by Pitches: A situation where the batter is struck by a pitched ball, and is awarded first base.
- Strikeouts: When the batter fails to hit the ball after three strikes, and is given an out.
- Hits: The total number of times a batter successfully reaches base by hitting the ball.

- Stolen Bases: The number of times a baserunner successfully advances to the next base while the pitcher is delivering a pitch.
- Grounded into Double Plays: The number of times a batter hits the ball on the ground, resulting in a double play.
- Caught Stealing: The number of times a baserunner attempts to steal a base, but is thrown out.
- Outs: The total number of outs recorded during a player's plate appearances. Includes strikeouts, flyouts, groundouts, etc.

Here's how we got the counts for each batting statistic:

- Plate Appearances (PA): Given*
- At Bat (AB): Given
- Walks (BB): Given
- Singles (1B): $PA - 2B - 3B - HR$
- Doubles (2B): Given
- Triples (3B): Given
- Home Runs (HR): Given
- Hit by Pitches (HBP): Given
- Strikeouts (SO): Given
- Hits (H): Given
- Stolen Bases (SB): Given
- Grounded into Double Plays (GIDP): Given
- Caught Stealing (CS): Given
- Outs (O): $PA - H - SO - BB - HBP$
- Outs - GIDP: $O - GIDP$

*Given means statistics directly given through our sources cited.

Here are our calculations to calculate the probability of each batting statistic, in terms of plate appearances:

- Plate Appearances (PA) = Plate Appearances (PA)

- At Bat (AB) = At Bat (AB)
- Walks (pBB) = BB / PA
- Singles (p1B) = 1B / PA
- Doubles (p2B) = 2B / PA
- Triples (p3B) = 3B / PA
- Home Runs (pHR) = HR / PA
- Hit by Pitches (pHBP) = HBP / PA
- Strikeouts (pSO) = SO / PA
- Hits (pH) = H / PA
- Stolen Bases (pSB) = SB / PA
- Grounded into Double Plays (pGIDP) = GIDP / PA
- Caught Stealing (pCS) = CS / PA
- Outs (pOUT) = OUT / PA

Here are some of our modeling assumptions:

- Conservative baserunning: runners advance as many bases as the batter does
- No errors: no misplays that let batters/runners reach or take extra bases
- No tagging up: runners can't advance on an out
- Double plays: must be force outs; will always occur at the furthest base runners, EXCEPT for 0 outs, (123)
- No triple plays
- Assume "Force Out.": (Fielder's choice) the "force" out will always be for the furthest base runner
- Stealing: Stolen bases are only considered when second base is open and there is a runner on first. If a stolen base is attempted, the player is always successful
- Memoryless: no previous games affect future games, no previous at bats affect future at bats
- Context neutral: probability of a given outcome is constant regardless of baserunners or outs

Techniques / Tools Used

To run our model, we performed Markov Chain analysis by building upon the code in the Colab Notebook given to us on Canvas. In Google Sheets, we created two 25×25 probability transition matrices (one for each team), with the states describing the base runner positions and number of outs at the beginning of each plate appearance. Since there are 8 possibilities for the bases to be occupied, and this can happen at 0, 1, or 2 outs, this gives $8 \times 3 = 24$ states. The final 25th state comes from the absorbing state of 3 outs. We then imported these matrices into our Google Colab notebook as NumPy arrays that we could work with for modeling.

To run the simulation, we built a function that simulates each half-inning individually. Each half-inning begins at state “---” (no outs, no baserunners) and uses the transition matrix to randomly select the next state. The model keeps track of how many runs are scored using a separate “run matrix,” and it accumulates these runs until it reaches state 24, which is the absorbing 3-out state. We then built another function that simulates a full 9-inning game by repeatedly calling the half-inning simulation for each team. Our final function simulates a three-game series by building on the previous functions and outputting the series results.

We first inputted our two matrices from Excel into NumPy arrays so that our Python code could interpret them—these represented our probability transition matrices, one for the Orioles and one for the Cubs. With the assistance of Claude AI, we then worked to clearly explain two important components of our model: the **run matrix** and the **random number generator**.

For the run matrix, our code generates a 25×25 matrix that mirrors the structure of the transition matrices. However, instead of probabilities, it contains the number of runs (from 0 to 4) scored for each possible transition between base–out states. This matrix is important because it allows the model to keep a running tally of runs once a transition has already been chosen.

Our code also includes a random number generator that converts probabilities into cumulative form over the interval from 0 to 1. This interval is partitioned according to the size of each non-zero probability in a given row of the transition matrix. Instead of deterministically choosing the most likely outcome, the model generates a random but uniform number between 0 and 1 (for

example, 0.45). Like a spinner, this number lands in one of the probability intervals, which determines the transition state that occurs.

When we actually simulate a half-inning (for example, starting with the Orioles batting), the model begins at state 0, meaning 0 outs and no baserunners. Suppose the simulation moves to state 8, which represents 1 out and no baserunners. The model then looks at the 8th row of the transition matrix and maps all non-zero probabilities onto the interval from 0 to 1 based on their relative sizes. A random number is generated and mapped onto this interval. If the number lands within the range corresponding to a probability of 0.0317, this outcome represents a home run. The model then records the transition from state i to state j , and this new state becomes the starting point for the next iteration.

Before continuing, the model checks the entry $[i,j]$ in the run matrix to determine how many runs scored on that transition. In this case, the value is 1, which makes sense since a home run with no baserunners scores one run. The model keeps a running total of runs and continues looping through this process. The simulation repeats until it reaches the absorbing state (state 24), which represents three outs and signals that the half-inning is over.

This process represents one simulated half-inning for the Orioles. The same half-inning simulation is then run for the Cubs. Running this process nine times for each team produces a full game, and comparing the final scores determines the winner. Repeating the full game simulation three times produces a three-game series, where the team with the majority of wins is declared the series winner.

Our model's construction (and frankly, parts of the interpretation) was assisted by help from generative AI. We also used AI to help create a Monte Carlo Simulation of 10,000 games between the Cubs and Orioles just to see who would be favored to win in the long run. The simulations outputted the average number of runs per game for each team, as well as the win percentages for each team across all 10,000 games.

Results

When running the Markov Chain three-game series simulation 10 times, we found that the Cubs won the series 6 times, and the Orioles won 4 times. Here's a summary of all 10 series simulations (first 3 series are shown in our presentation in more detail):

Series 1 Winner: Cubs (2-1)

Series 2 Winner: Orioles (2-1)

Series 3 Winner: Cubs (3-0)

Series 4 Winner: Cubs (3-0)

Series 5 Winner: Orioles (3-0)

Series 6 Winner: Cubs (2-1)

Series 7 Winner: Cubs (2-1)

Series 8 Winner: Orioles (3-0)

Series 9 Winner: Orioles (2-1)

Series 10 Winner: Cubs (3-0)

When running the Monte Carlo simulation of 10,000 games, we found that the Cubs averaged 3.89 runs per game with a win percentage of 50.44%, while the Orioles averaged 3.31 runs per game with a win percentage of 38.60%. The rest of the 10.96% of games resulted in ties between the two teams. Our code for each individual game does not include ties, but instead goes to extra innings. However, for this simulation we did not want these extra innings to skew the average runs per game for either team, so we decided to make ties a possible outcome.

Both of these simulations show that the Chicago Cubs would typically win in a three-game, 9-inning series over the Baltimore Orioles, based on regular season batting statistics for 2025.

Interpretation

Our results of the Cubs being favored to win the three-game, 9-inning series over the Orioles are consistent with our expectations of what our model would output, given both teams' season records and batting statistics over the course of the 2025 regular season. The Cubs had a winning record of 92-70, while the Orioles had a losing record of 75-87. With a relatively large sample

size of 162 games, it already seemed likely that this could indicate the Cubs being the stronger team, although not by a huge margin. Per plate appearance, the Cubs had slightly higher probabilities of walks, singles, doubles, triples, hits, stolen bases, and home runs. Furthermore, the Cubs had lower probabilities of strikeouts, grounded into double plays, and outs. Each of these factors were implemented into our model and played a part in our results ultimately showing the Cubs being favored over the Orioles.

Assessment

Overall, the model performed well in giving results that seem to reflect 2025 batting statistics. Like mentioned in the “Interpretation” section, prior to modeling we had expected the Cubs to have the advantage over the Orioles based on their overall record and team batting statistics, and our hypothesis was validated through our results. Despite this, our model still came with limitations which may have over-inflated or under-inflated runs, due in part to some assumptions that were made. One of these assumptions includes the steal sequence, where we assume that a player cannot be caught stealing a base; in other words, whenever a player attempts to steal a base, they successfully do so. Another limiting assumption was conservative baserunning, which could potentially underestimate both runs and outs. Finally, our model also does not take into consideration some real-game situations such as playing at home vs. away, or having injuries/fatigue. Moving forward, it would be interesting to alter our model and address some of these limitations, and compare those results to our current ones.

Sources

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